Make Your Network Secure with PCAP and Snort

PCAP is an application programming interface (API) for capturing network traffic (packets). Snort is a tool for detecting network intrusion. This article explains how they can be used in tandem to analyse network traffic and detect any attacks on the network.

Packet capture is a classic, frequently performed task carried out by network administrators. This is done to detect any suspicious activity in the network. Any out-of-the-way or abnormal activity is analysed by intrusion detection system (IDS) tools in order to classify the attack or the type of traffic. There are numerous IDS tools available, including open source products, that classify the attacks or traffic based on information gathered from the PCAP (packet capture) files fetched from honey pots or servers.

PCAP (packet capture)

Issues of intrusion in the network by different media can be tackled by making use of PCAP (packet capture), which has an application programming interface (API) for capturing network traffic from ports, IP addresses and associated parameters. In the case of UNIX-like systems, PCAP is implemented in the Libpcap library. In Windows, it is implemented through WinPcap, which is the Windows version of Libpcap.

The base API of PCAP is in the C programming language. To implement PCAP in other programming languages such as Java, .NET and Web-based scripting languages, a wrapper is used, but remember that neither Libpcap nor WinPcap provide these wrappers by default. In the case of C++, the programs can link directly to the C API or make use of an object-oriented wrapper.

The MIME type for the file format that is created and read by Libpcap is *application/vnd.tcpdump.pcap*. The classical file extension for PCAP is *.pcap*. In some tools, *.cap* and *.dmp* file extensions are also used.

Libpcap and WinPcap are associated, in terms of packet capturing as well as filtering engines, with many open source and commercial network tools. These include protocol analysers (packet sniffers), network investigators, network IDEs, traffic generators and network analysers.

A feature of Libpcap is that the captured files can be exported and saved to a file. A captured file that is saved in the format that Libpcap and WinPcap use can be easily analysed by applications that understand this format, including tcpdump, Wireshark, NetworkMiner and many others.

Tools for reading Libpcap

Here is a list of network analysis tools that make use of Libpcap.

- **Tcpdump:** A tool to capture and dump packets for forensics and investigation.
- **ngrep (Network Grep):** Shows packet data in a user-friendly output scenario.